

	Course Name: Advanced Web Technology	EXPERIMENT NO. 4
	Course Code: 20CP314P	Branch: CSE
	Faculty: Komal Singh	Semester: VI
Submitted by: Harsh Shah Roll no: 21BCP359		

Objective: Building models with Mongoose

Create a database schema and model using mongoose and perform MongoDB queries for operations such as insert, update, and delete while also establishing a connection to the MongoDB database.

Mongoose Connection

```
const mongoose = require("mongoose");

mongoose
  .connect("mongodb://localhost/mydb")
  .then(() => console.log("Connected to MongoDB successfully!"))
  .catch((error) => console.error("Error connecting to MongoDB", error));
```

```
(base) PS D:\DEV\Web Development\PDEU-AWT\Practical_4> npm start

> practical_4@1.0.0 start
> nodemon task1.js

[nodemon] 3.1.0
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node task1.js`
Connected to MongoDB successfully!
```

Database Schema

```
const schema = new mongoose.Schema({

  name: {
    type: String,
    required: true,
  },
  email: {
    type: String,
```

```
    required: true,
    unique: true,
  },
  age: {
    type: Number,
    default: 0,
  },
  deregistered: {
    type: Date,
    default: Date.now,
  },
});
const obj = mongoose.model("subsription", schema);
```

Insert Operation

```
const newobj = new obj({
  name: "Harsh Shah",
  email: "harsh.sce21@sot.pdpu.ac.in",
  age: 20,
});
newobj
  .save()
  .then((kri) => console.log("User created:", kri))
  .catch((err) => console.error(err));
```

```
Connected to MongoDB successfully!
User created: {
  name: 'Harsh Shah',
  email: 'harsh.sce21@sot.pdpu.ac.in',
  age: 20,
  _id: new ObjectId('660fd63ed60d5b00ccc41371'),
  deregistered: 2024-04-05T10:45:18.554Z,
  __v: 0
}
```

```
_id: ObjectId('660fd63ed60d5b00ccc41371')
name : "Harsh Shah"
email : "harsh.sce21@sot.pdpu.ac.in"
age : 20
deregistered : 2024-04-05T10:45:18.554+00:00
__v : 0
```

a. Insert One

```
const Newuser = new obj({ name: "krish", email: "ky@gmail.com", age: 20 });
```

```
obj.collection
```

```
.insertOne(Newuser)
```

```
.then((data) => {
```

```
  console.log(data);
```

```
})
```

```
.catch((err) => {
```

```
  console.log(err);
```

```
});
```

```
Connected to MongoDB successfully!
{
  acknowledged: true,
  insertedId: new ObjectId('660fd845ddcedbff28571b2e')
}
```

b. Insert Many

```
const Newuser = new obj({ name: "krishna", email: "ku@gmail.com", age: 20 });
```

```
const Newuser1 = new obj({ name: "kriti", email: "kr@gmail.com", age: 21 });
```

```
obj.collection
```

```
.insertMany([Newuser, Newuser1])
```

```
.then((data) => {
```

```
  console.log(data);
```

```
})
```

```
.catch((err) => {
```

```
  console.log(err);
```

```
});
```

```
Connected to MongoDB successfully!
{
  acknowledged: true,
  insertedCount: 2,
  insertedIds: {
    '0': new ObjectId('660fd8829d209dcbb3368d5b'),
    '1': new ObjectId('660fd8829d209dcbb3368d5c')
  }
}
```

Update Operation

a. Update One

```
obj.updateOne({ name: "krishna" }, { $set: { name: "madhav" } })  
  .then((docs) => {  
    if (docs) {  
      console.log(docs);  
    } else {  
      console.log("no such user exist");  
    }  
  })  
  .catch((err) => {  
    console.log(err);  
  });
```

```
Connected to MongoDB successfully!  
{  
  acknowledged: true,  
  modifiedCount: 1,  
  upsertedId: null,  
  upsertedCount: 0,  
  matchedCount: 1  
}
```

b. Update Many

```
obj.updateMany({age: 20},{ $set:{ age: 15}})  
  .then((docs)=>{  
    if(docs) {  
      console.log(docs);  
    } else {  
      console.log("no such user exist");  
    }  
  })  
  .catch((err)=>{  
    console.log(err);  
  });
```

```
Connected to MongoDB successfully!  
{  
  acknowledged: true,  
  modifiedCount: 4,  
  upsertedId: null,  
  upsertedCount: 0,  
  matchedCount: 4  
}
```

Find/Search Operation

a. Find without query

```
obj.find({})  
  
  .then((doc) => {  
    console.log(doc);  
  })  
  
  .catch((err) => {  
    console.log(err);  
  });
```

```
Connected to MongoDB successfully!  
[  
  {  
    _id: new ObjectId('660fd63ed60d5b00ccc41371'),  
    name: 'Harsh Shah',  
    email: 'harsh.sce21@sot.pdpu.ac.in',  
    age: 15,  
    deregistered: 2024-04-05T10:45:18.554Z,  
    __v: 0  
  },  
  {  
    _id: new ObjectId('660fd845ddcedbff28571b2e'),  
    name: 'krish',  
    email: 'ky@gmail.com',  
    age: 15,  
    deregistered: 2024-04-05T10:53:57.063Z  
  },  
  {  
    _id: new ObjectId('660fd8829d209dcbb3368d5b'),  
    name: 'madhav',  
    email: 'ku@gmail.com',  
    age: 15,  
    deregistered: 2024-04-05T10:54:58.784Z  
  },  
  {  
    _id: new ObjectId('660fd8829d209dcbb3368d5c'),  
    name: 'kriti',  
    email: 'kr@gmail.com',  
    age: 21,  
    deregistered: 2024-04-05T10:54:58.784Z  
  },  
  {  
    _id: new ObjectId('660fd92a27c646b3dd442ec0'),  
    name: 'dev',  
    email: 'dev@gmail.com',  
    age: 15,  
    deregistered: 2024-04-05T10:57:46.699Z  
  }  
]
```

b. Find with query

```
obj.find({ name: "dev" })  
  
  .then((doc) => {  
    console.log(doc);  
  })  
  
  .catch((err) => {  
    console.log(err);  
  });
```

```
Connected to MongoDB successfully!
[
  {
    _id: new ObjectId('660fd92a27c646b3dd442ec0'),
    name: 'dev',
    email: 'dev@gmail.com',
    age: 15,
    deregistered: 2024-04-05T10:57:46.699Z
  }
]
```

Delete Operation

a. Delete One

```
obj.deleteOne({name: "kriti"})
.then((docs)=>{
  if(docs) {
    console.log(docs);
  } else {
    console.log("no such user exist");
  }
}).catch((err)=>{
  console.log(err);
});
```

```
Connected to MongoDB successfully!
{ acknowledged: true, deletedCount: 1 }
```

b. Delete Many

```
obj.deleteMany({age: 21})
.then((docs)=>{
  if(docs) {
    console.log(docs);
  } else {
    console.log("no such user exist");
  }
}).catch((err)=>{
  console.log(err);
});
```

```
Connected to MongoDB successfully!
{ acknowledged: true, deletedCount: 4 }
```

MongoDB Compass

mydb.subscriptions

4DOCUMENTS2INDEXES

Documents

Aggregations

Schema

Indexes

Validation

Filter

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

</>

Options

ADD DATA

EXPORT DATA

UPDATE

DELETE

1 - 4 of 4

```
{
  "_id": ObjectId("660fd63ed60d5b00ccc41371"),
  "name": "Harsh Shah",
  "email": "harsh.sce21@sot.pdpu.ac.in",
  "age": 15,
  "deregistered": 2024-04-05T10:45:18.554+00:00,
  "__v": 0
}
```

```
{
  "_id": ObjectId("660fd845ddcedbff28571b2e"),
  "name": "krish",
  "email": "ky@gmail.com",
  "age": 15,
  "deregistered": 2024-04-05T10:53:57.063+00:00
}
```

```
{
  "_id": ObjectId("660fd8829d209dcbb3368d5b"),
  "name": "madhav",
  "email": "ku@gmail.com",
  "age": 15,
  "deregistered": 2024-04-05T10:54:58.784+00:00
}
```

My Queries

mydb

mydb

subscriptions

mydb.subscriptions

0DOCUMENTS2INDEXES

Documents

Aggregations

Schema

Indexes

Validation

Filter

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

</>

Options

ADD DATA

EXPORT DATA

UPDATE

DELETE

0 - 0 of 0



This collection has no data

It only takes a few seconds to import data from a JSON or CSV file.

Import Data